Practice Problem Set 2

ECON 320 – Introduction to Mathematical Economics

Fall 2025

1. Consider the system:
$$\begin{cases} x + y = \alpha \\ x - y = \beta \end{cases}$$
   (a) Solve for $x(\alpha, \beta)$ and $y(\alpha, \beta)$.
   (b) Compute $\dfrac{\partial x}{\partial \alpha}, \dfrac{\partial x}{\partial \beta}, \dfrac{\partial y}{\partial \alpha}, \dfrac{\partial y}{\partial \beta}$.
   (c) Interpret the effect of α and β on the equilibrium values of x and y.

2. Let the implicit function be defined by:
$$F(x, y, p) = x^2 + xy + y^2 - p = 0$$
   (a) Use total differentiation to compute $\dfrac{\mathrm{d}x}{\mathrm{d}p}$ and $\dfrac{\mathrm{d}y}{\mathrm{d}p}$.
   (b) At the point $(x, y) = (1, 1), p = 3$, determine the signs of the comparative static effects.

3. Consider the nonlinear system:
$$\begin{cases} x^2 + y = p \\ x + y^2 = q \end{cases}$$
   (a) Write the Jacobian matrix $J(x, y)$.
   (b) Compute $\det(J(x, y))$.
   (c) State the condition under which the system has a locally unique solution $(x(p, q), y(p, q))$.

4. The system is:
$$\begin{cases} F_1(x, y, \theta) = x^2 + y - \theta = 0 \\ F_2(x, y, \theta) = xy - 1 = 0 \end{cases}$$
   (a) Compute the Jacobian matrix with respect to (x, y).
   (b) Show that the determinant is nonzero at $(x, y) = (1, 1), \theta = 2$.
   (c) Apply the Implicit Function Theorem to argue about existence and differentiability of $(x(\theta), y(\theta))$.

(d) Compute $\frac{dx}{d\theta}, \frac{dy}{d\theta}$ at $\theta = 2$.

5. Suppose:

$$\begin{cases} x + y + z = \theta \\ x^2 + y^2 = 1 \\ z^2 = x \end{cases}$$

(a) Write the Jacobian matrix of the system with respect to (x, y, z).

(b) Use Cramer's Rule to find $\frac{\partial(x, y, z)}{\partial\theta}$.

(c) Discuss the effect of a marginal change in θ on the three variables.

6. Let the transformation be:

$$u = x^2 + y^2, \quad v = xy$$

(a) Compute the Jacobian $\frac{\partial(u, v)}{\partial(x, y)}$.

(b) If possible, compute the inverse Jacobian $\frac{\partial(x, y)}{\partial(u, v)}$.

(c) Explain how this transformation could simplify optimization problems with quadratic forms.